William Wei

wjw9857@nyu.edu | (437)-445-3268 | Ajax, Ontario
github.com/william-wei0 | linkedin.com/in/williamwei0/

EDUCATION

New York University **May 2025**
Bachelor of Science in Biomolecular Science GPA: 3.95/4.00

- **Minors:** Computer Science and Mathematics.
- **Honors/Awards**: Roland Ward Award, Summa Cum Laude, Dean's List.

Relevant Courses: Object Oriented Programming, Data Structures & Algorithms, Digital Logic and State Machine Design, Linear Algebra & Differential Equations, Calculus I-III, Statistics, Physics I-III, Modern & Solid-State Physics.

EXPERIENCE

Machine Learning Research Assistant **Sept 2024 – Present**
New York University, Smart Microphysiological Systems for Precision Medicine New York, NY

- Designed an end-to-end data analytics workflow with an Attention-LSTM model (Pytorch) to forecast 3-year pancreatic cancer responses, generating discrete-time survival reports & patient health scores with AUC of 0.89.
- Developed a scalable image-to-data pipeline with Python/OpenCV to preprocess & extract 15 features from 80k+ time-lapse images, producing clean NumPy/Pandas datasets with 900k+ cell instances.
- Built data analysis pipelines to perform log-rank tests & calculate p-values with SciPy/scikit, explore embeddings of T-cell behavior with k-means clustering/UMAP, and generate KM-curves with Matplotlib for publications.

Physical Chemistry I Course Assistant **Jan 2025 – May 2025**
New York University, Department of Chemical and Biomolecular Engineering New York, NY

- Developed a full stack website (physicalchem.org) using React/Express.js/PostgreSQL/TypeScript with interactive 3D simulations of quantum mechanics concepts to help 100+ students learn physical chemistry concepts.
- Designed and implemented a PostgreSQL relational database with Prisma ORM to store/retrieve data from 100 user accounts, and tracking student progress and student preferences in simulations.
- Built and secured a RESTful API with Node.js and Express.js, implementing role-based access control, JSON Web Token-based authentication, and CORS policies to restrict cross-origin requests.
- Designed a responsive and interactive UI for simulations using React/Tailwind/Vite.
- Automated testing and deployment with a CI/CD pipeline using Vitest and GitHub Actions.
- Deployed using AWS cloud hosting with Amplify/Elastic Beanstalk/Amazon RDS, improving up-time by 5%.

Computer Vision Research Assistant **Sept 2024 – May 2025**
New York University, Garetz Laboratory New York, NY

- Developed and fine-tuned a YOLO-based object tracking model to automate measurements of crystal size/shape from images, reducing data acquisition times by 32% and an accuracy within 15% of expected values.
- Dockerized models and inference workflows to ensure consistent deployments and results across environments.
- Maintained documentation on Github to support cross-functionality between developers and clients.

PROJECTS

2D Tile-Based Game Engine (C++/OpenGL) | Source Code **Jan 2025 – May 2025**

- Built a 2D game engine in C++/OpenGL with tile rendering, dynamic entities, and multi-scene state management.
- Applied inheritance and polymorphism to share behavior across player, enemy, and projectile entity types.
- Abstracted tile collision exposing a clean interface for collision detection and penetration resolution.
- Encapsulated GLSL shader compilation and GPU resource cleanup into a reusable ShaderProgram class.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, SQL, C++, HTML, CSS.
- **Libraires:** PyTorch, NumPy, Pandas, SciPy, scikit-learn, Matplotlib, OpenCV, ThreeJS. Vitest
- **Frameworks:** React, Node.js, Express.js, PostgreSQL, Tailwind, Docker, Git, Github Actions, AWS RDS